



MOHAMMED BENGHANEM

AI ENGINEER | FULL STACK DEVELOPER

Available immediately | On-site, hybrid or remote | Open to relocation

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SUMMARY

AI Engineer specializing in generative AI and full-stack web development. Experienced in designing RAG systems, fine-tuning large language models (LLMs) with QLoRA, and orchestrating AI agents. Proficient in Python, FastAPI, and React/Next.js for integrating predictive models and deploying secure applications.

EXPERIENCE

Freelance Web and AI Developer | Taxi Tremplin

Aug 2026 – Present

- Designed the complete backend using NestJS, PostgreSQL/PostGIS, and Redis, including location-based driver dispatch, fare calculation, SMS authentication, and notifications. Deployed using Docker.
- Built a React/TypeScript administration dashboard to manage drivers, customers, rides, fares, and payments.
- Developed an intelligent booking assistant powered by the Claude API (Anthropic), enabling users to book taxis in natural language through AI agents and tool calling.

AI and Web Development Intern | Yosobox

Mar - sep 2026

- Developed two moisture content (LoD) prediction models: a physics-based model and an artificial neural network.
- Integrated both models into a real-time dashboard using a FastAPI microservice, a secure Node.js/Express API, and Supabase storage.
- Created a secure API endpoint with admin API key authentication and rate limiting to trigger ANN predictions.
- Fixed integration bugs and made the physics-based model's drying curve

AI and Web Development Intern | Lotus Capital

Jan – Mar 2026

- Developed a social media competitive intelligence platform featuring multi-source data collection from X, LinkedIn, RSS, and Reddit, along with engagement analysis.
- Designed multi-agent workflows using LangGraph to detect trends and generate content, with human-in-the-loop validation at each stage.
- Implemented embedding-based similarity search and duplicate detection using pgvector, comparing published posts with competitors' content.

PROJECTS

ClauseLens - Contract Clause Analysis with RAG and QLoRA Fine-Tuning |, PyTorch, FastAPI, PostgreSQL, Next.js

- Built a RAG application that analyzes commercial contracts against a checklist of 12 legal clauses, producing structured responses supported by exact citations verified on the server.
- Implemented hybrid semantic and full-text search using PostgreSQL/pgvector, selected after comparison on a validation set; conducted leakage-free evaluation on 510 contracts annotated by legal professionals.
- Fine-tuned Qwen2.5-1.5B using QLoRA and Hugging Face PEFT/TRL on a GPU with 6 GB of VRAM, reducing the false-positive rate from 82% to 0.1% and increasing the verified citation rate from 69% to 95% on previously unseen contracts.

EDUCATION

Master's Degree in Big Data and Artificial Intelligence | Faculty of Sciences, Ibn Tofail University | 2024 - 2026

Bachelor's Degree in Mathematics and Computer Science | Faculty of Sciences, Ibn Tofail University | 2021 - 2024

TECHNICAL SKILLS

Backend / Databases: Python, JavaScript, Node.js, Express.js, FastAPI, PostgreSQL, MongoDB, ChromaDB, Supabase

Machine Learning & NLP: scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, fine-tuning, LLMs, RAG, prompt engineering

Frontend: React, TypeScript, Next.js, Tailwind CSS

DevOps / Tools : Docker, Git, Bash, AWS

CERTIFICATIONS & LANGUAGES

Certifications: AWS Cloud Technical Essentials, Machine Learning with Python - IBM, Tools for Data Science - IBM

Languages: Arabic (native) | English (fluent, C1) | French (professional)